

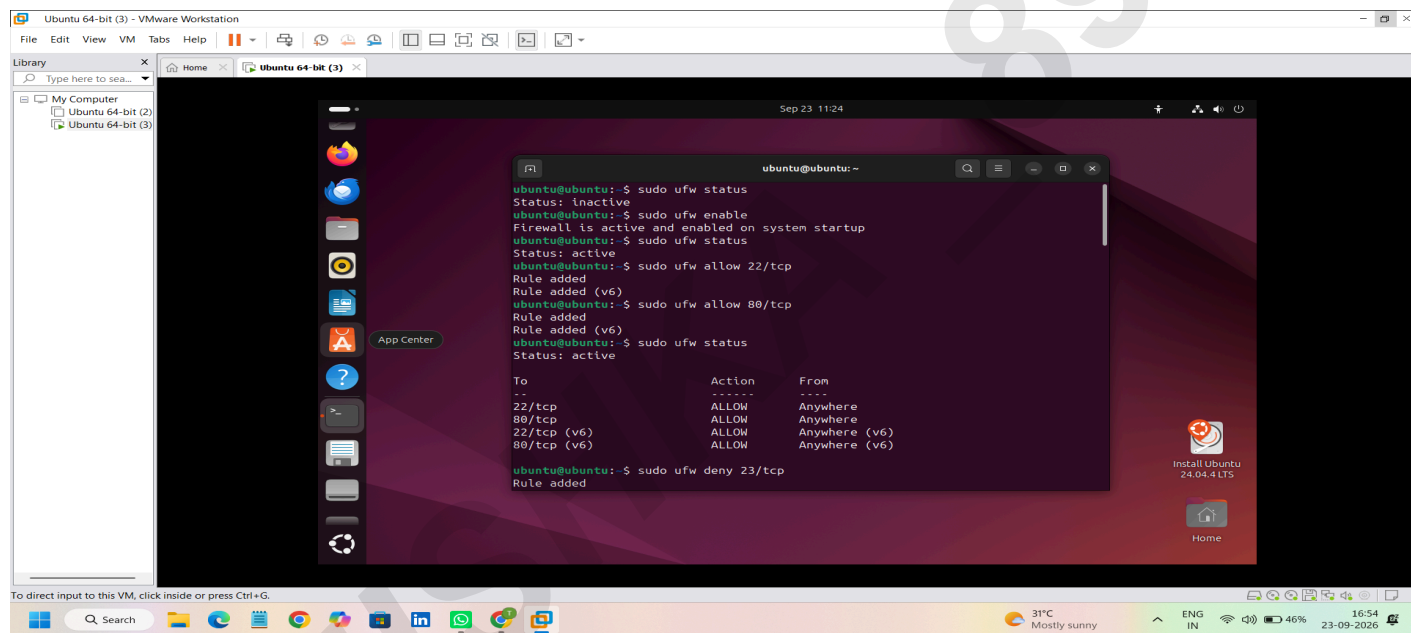
SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: CYBER SECURITY

PRACTICAL NO: 10

Aim:- Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Step 1: Open Ubuntu Terminal



```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

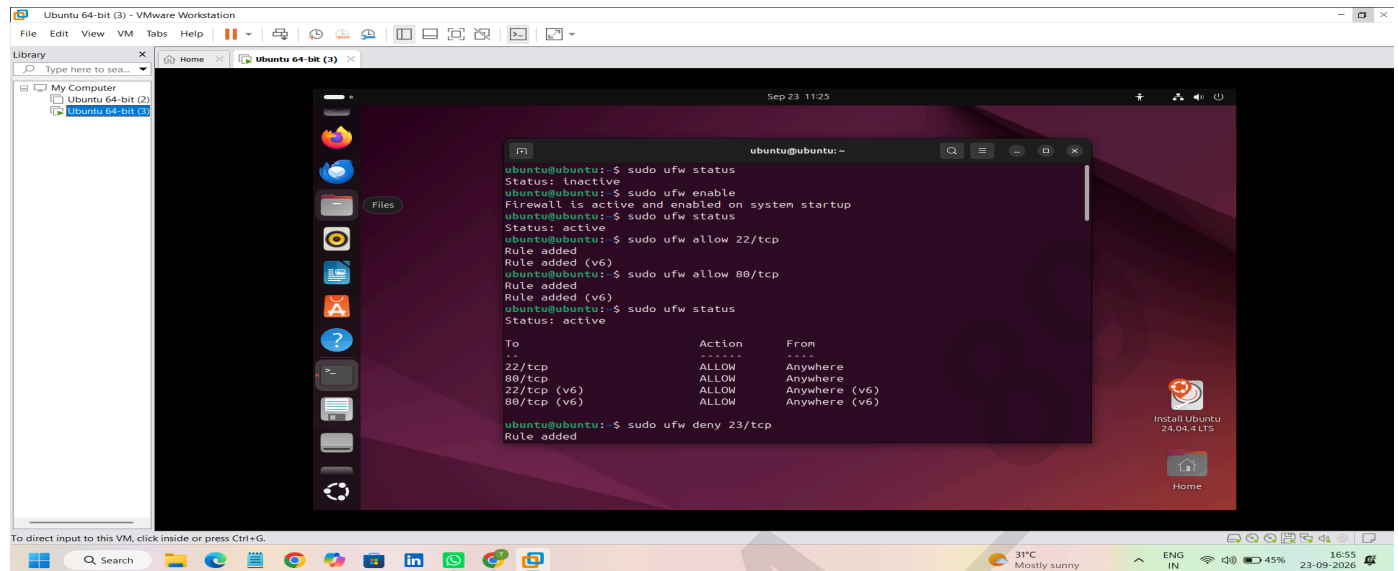
To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

Step 2: Enable the Firewall

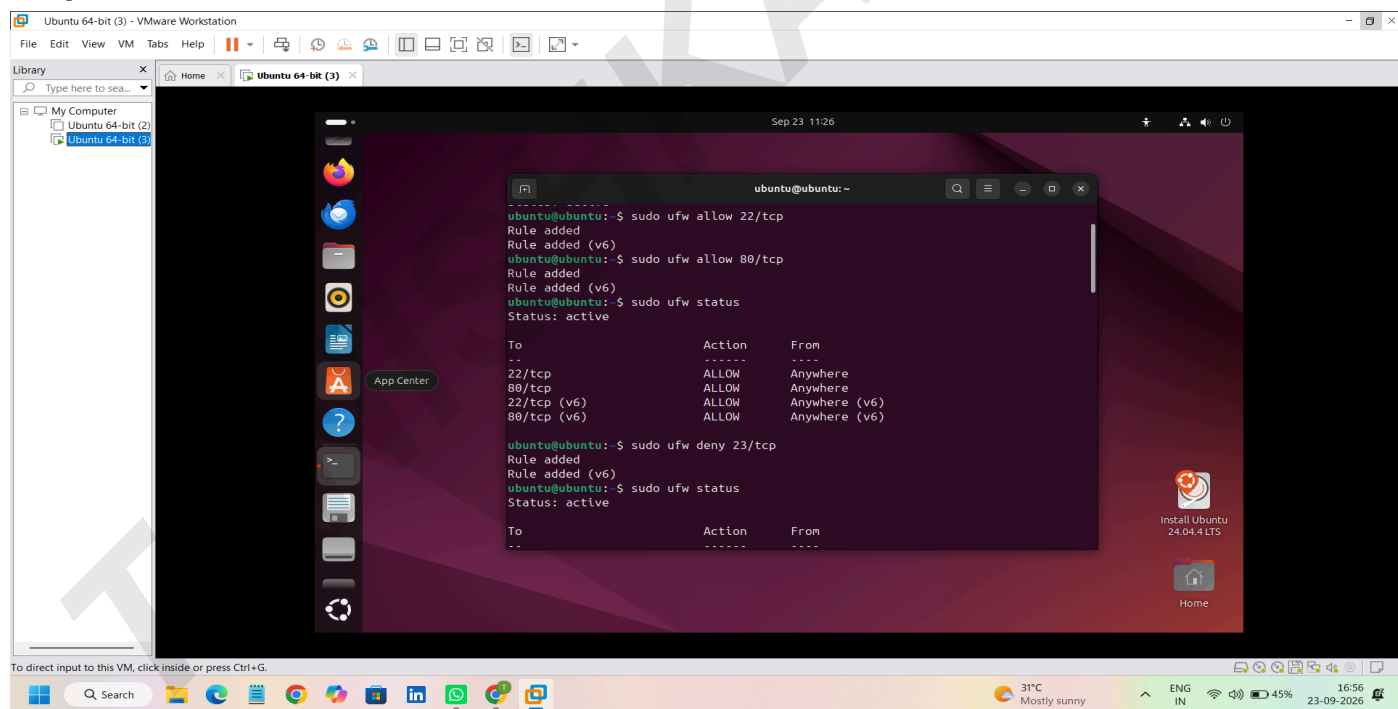
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Step 3: Allow SSH Traffic



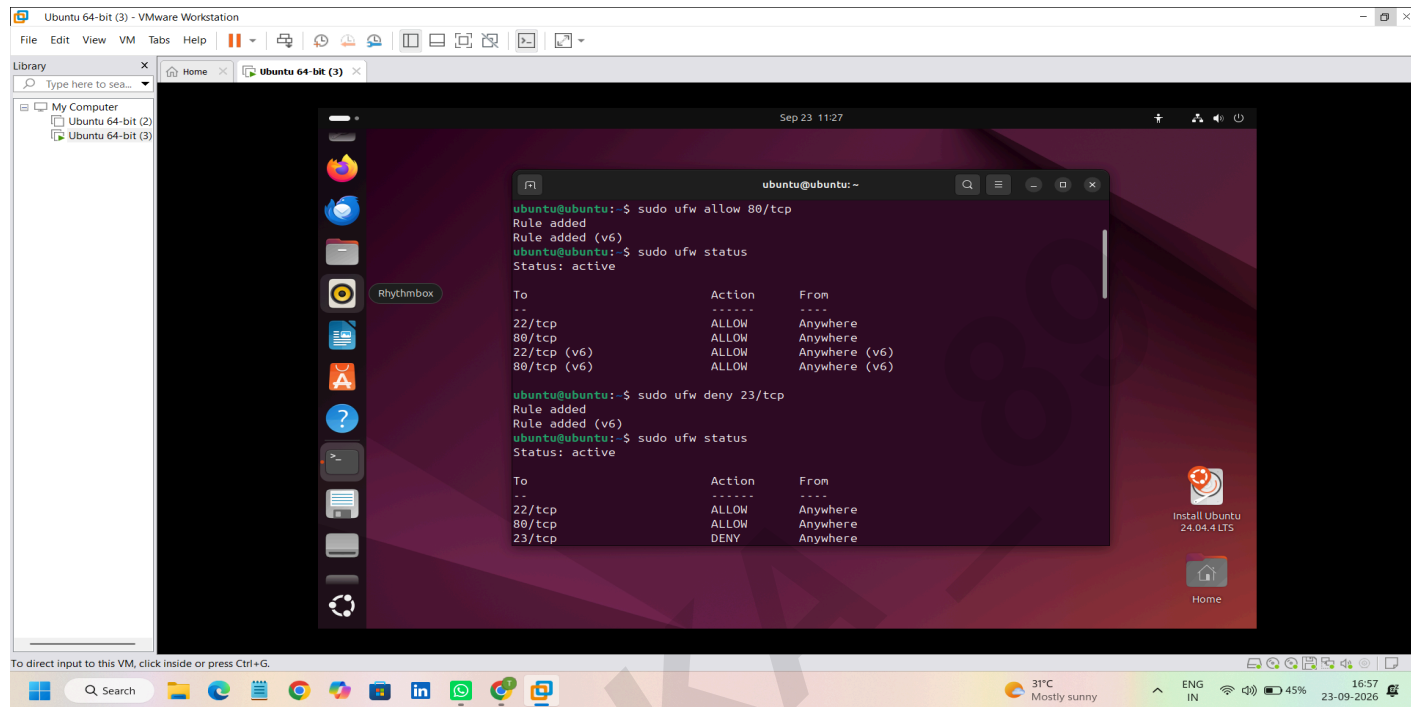
Step 4:- Step 4: Allow HTTP Traffic

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ROLL NO: T089

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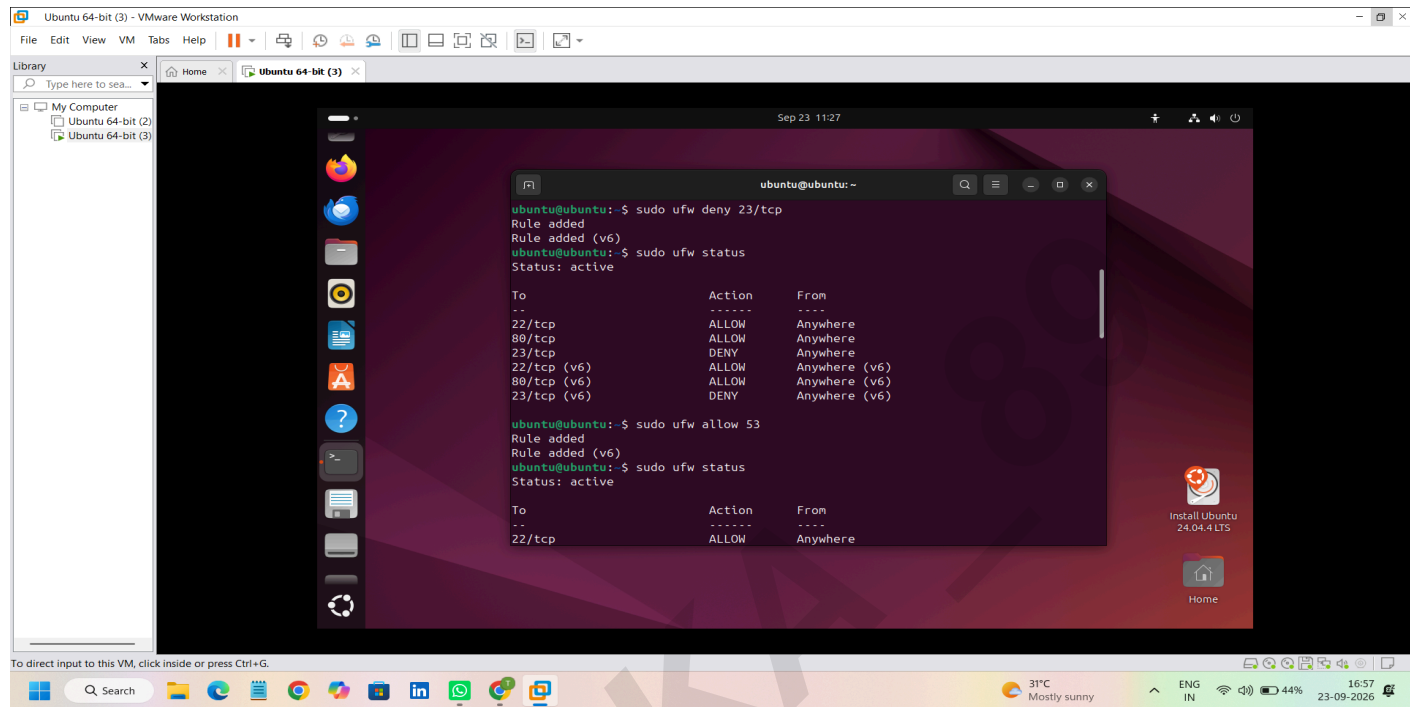
Step 5: Block a Port

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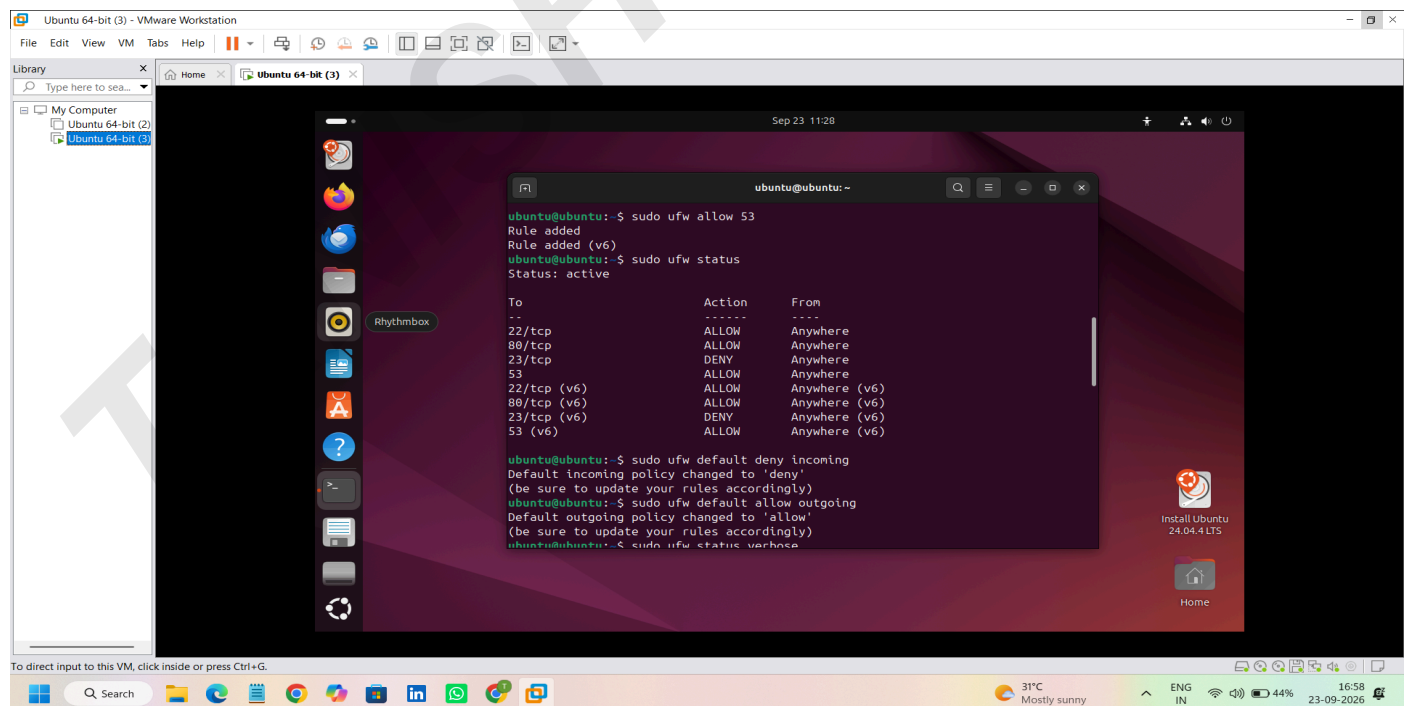
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Step 6: Allow DNS Traffic



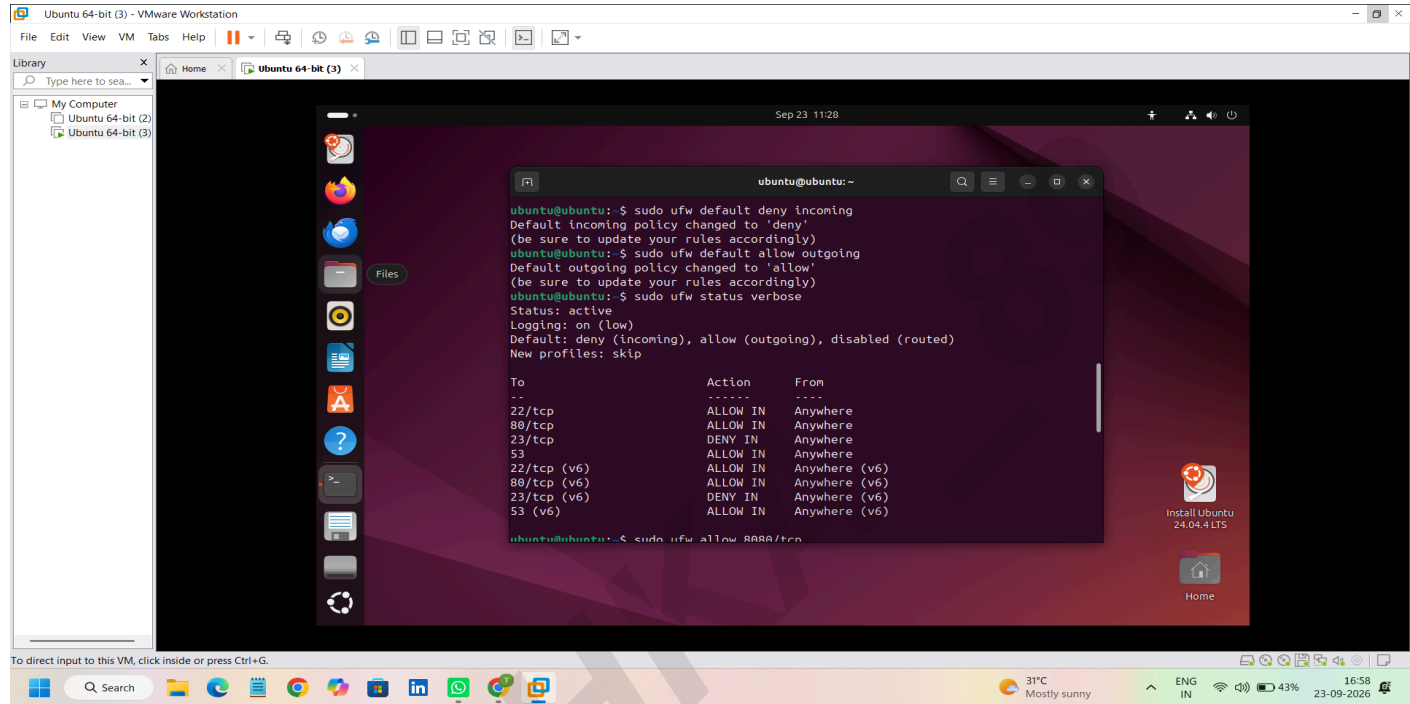
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Step 7: Set Default Firewall Policies



```
ubuntu@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
ubuntu@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
ubuntu@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

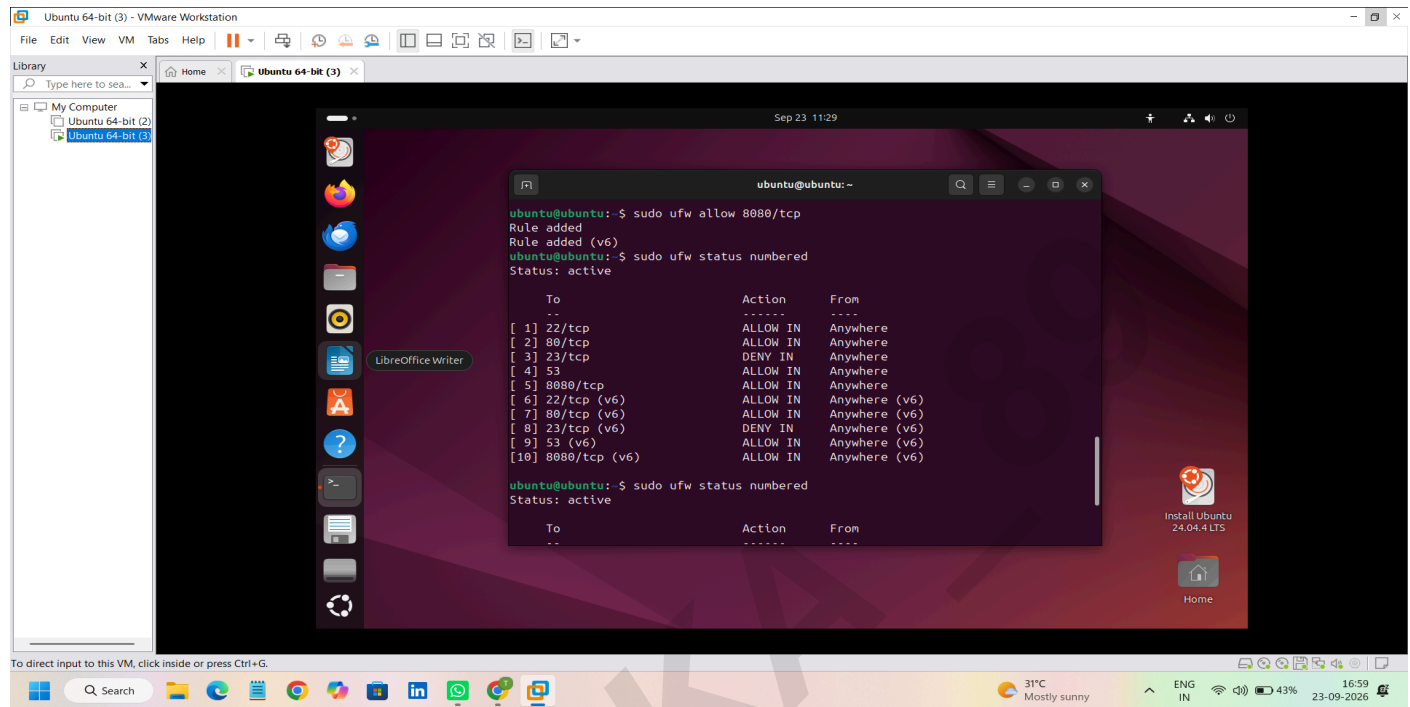
To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
23/tcp DENY IN Anywhere
53 ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
23/tcp (v6) DENY IN Anywhere (v6)
53 (v6) ALLOW IN Anywhere (v6)
```

Step 8: Create a Rule for a Specific Port

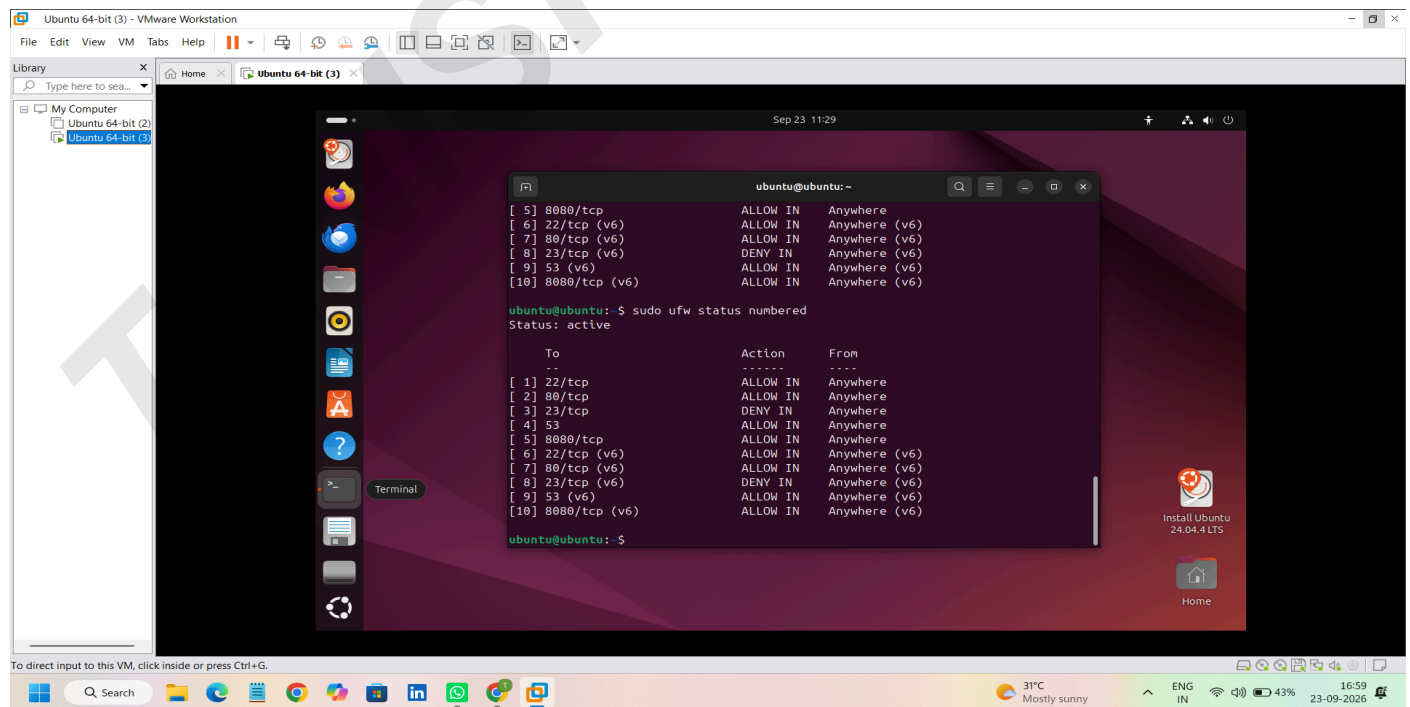
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Step 9: Test the Firewall



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SUBJECT: CYBER SECURITY

PRACTICAL NO: 10

To	Action	From
[1] 22/tcp	ALLOW IN	Anywhere
[2] 80/tcp	ALLOW IN	Anywhere
[3] 23/tcp	DENY IN	Anywhere
[4] 53	ALLOW IN	Anywhere
[5] 8080/tcp	ALLOW IN	Anywhere
[6] 22/tcp (v6)	ALLOW IN	Anywhere (v6)
[7] 80/tcp (v6)	ALLOW IN	Anywhere (v6)
[8] 23/tcp (v6)	DENY IN	Anywhere (v6)
[9] 53 (v6)	ALLOW IN	Anywhere (v6)
[10] 8080/tcp (v6)	ALLOW IN	Anywhere (v6)

ubuntu@ubuntu:~\$ nc -zv 127.0.0.1 8080
nc: connect to 127.0.0.1 port 8080 (tcp) failed: Connection refused
ubuntu@ubuntu:~\$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
Keyboard interrupt received, exiting.
ubuntu@ubuntu:~\$

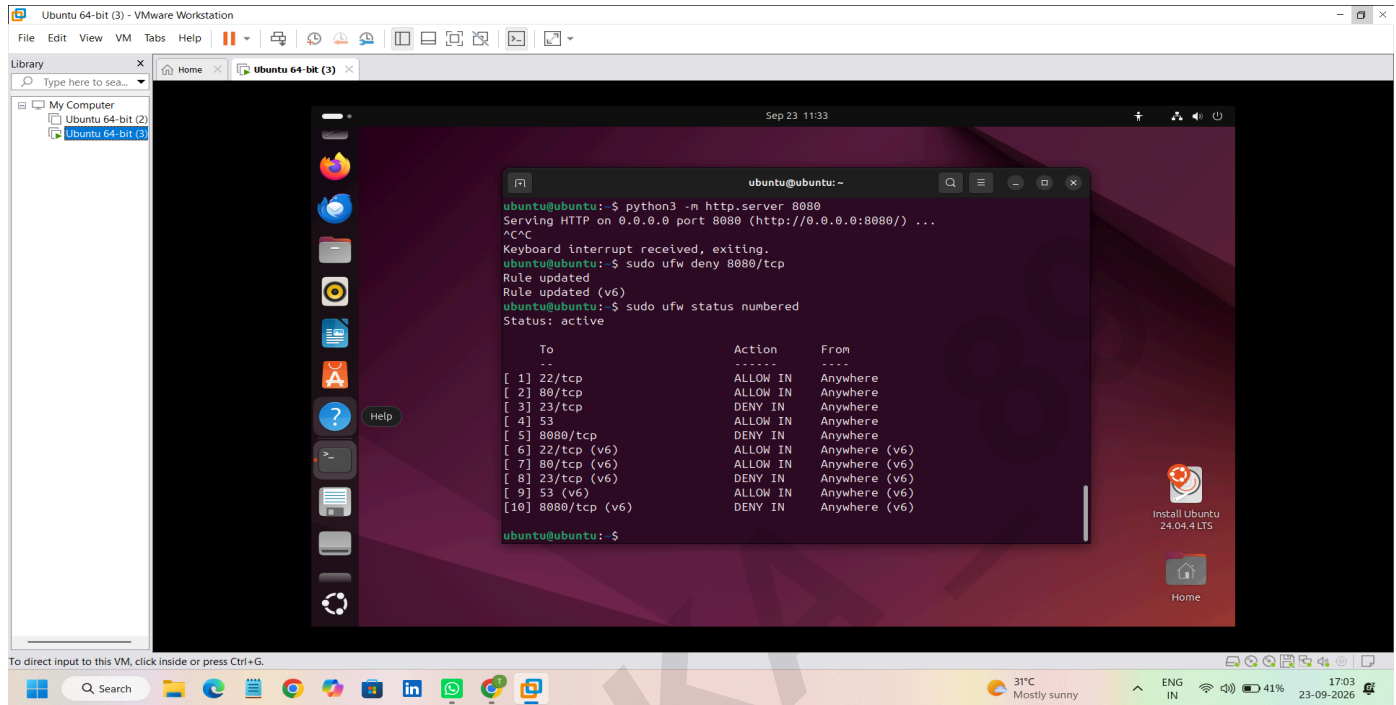
Step 10: Test Blocking

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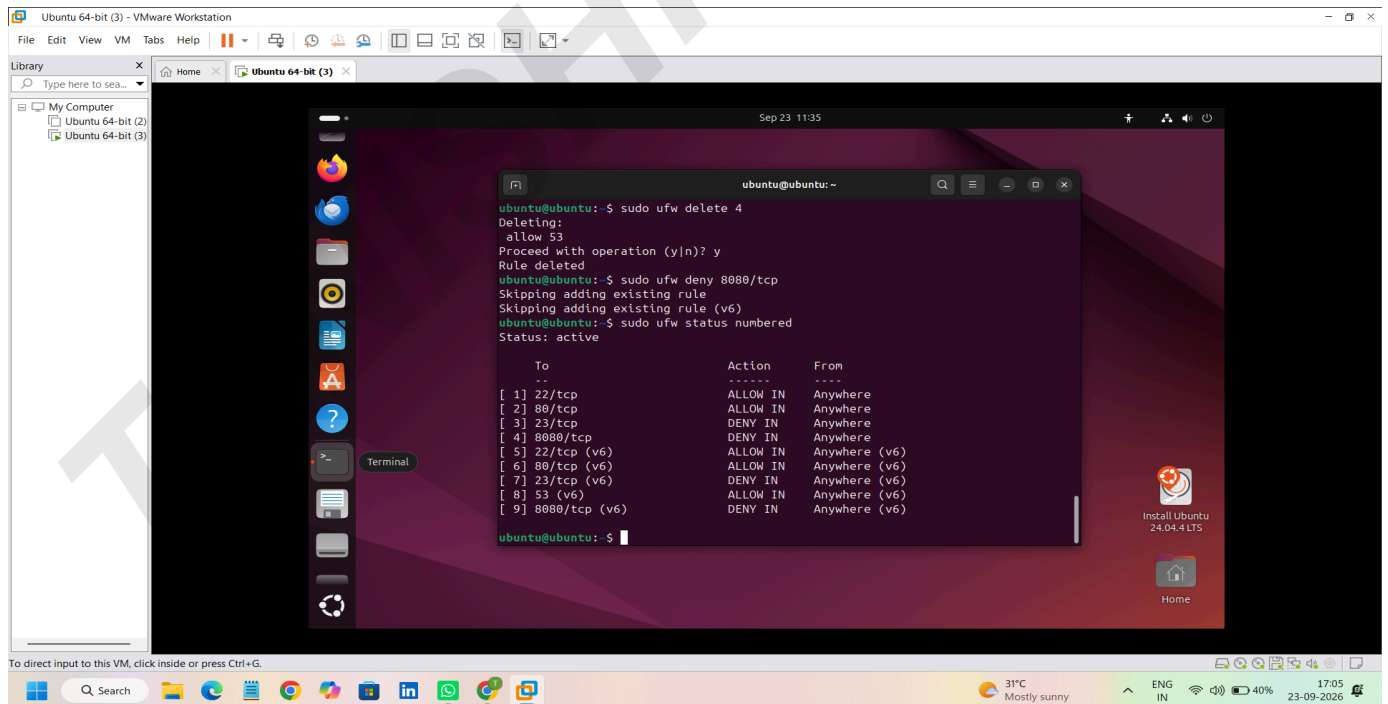
PRACTICAL NO: 10



```
ubuntu@ubuntu:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
^C^C
Keyboard interrupt received, exiting.
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp
Rule updated
Rule updated (v6)
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 23/tcp DENY IN Anywhere
[ 4] 53 ALLOW IN Anywhere
[ 5] 8080/tcp DENY IN Anywhere
[ 6] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 8] 23/tcp (v6) DENY IN Anywhere (v6)
[ 9] 53 (v6) ALLOW IN Anywhere (v6)
[10] 8080/tcp (v6) DENY IN Anywhere (v6)

ubuntu@ubuntu:~$
```



```
ubuntu@ubuntu:~$ sudo ufw delete 4
Deleting:
allow 53
Proceed with operation (y/n)? y
Rule deleted
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 23/tcp DENY IN Anywhere
[ 4] 8080/tcp DENY IN Anywhere
[ 5] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 6] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 23/tcp (v6) DENY IN Anywhere (v6)
[ 8] 53 (v6) ALLOW IN Anywhere (v6)
[ 9] 8080/tcp (v6) DENY IN Anywhere (v6)

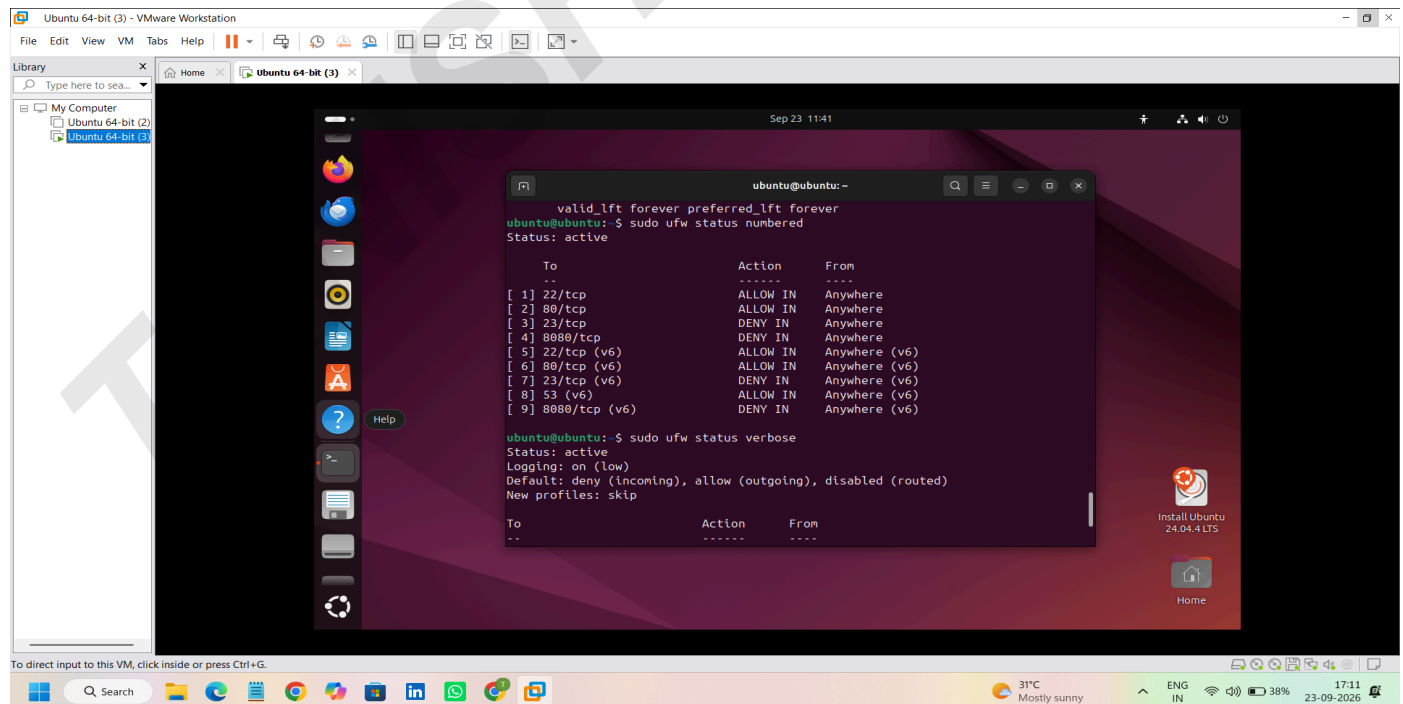
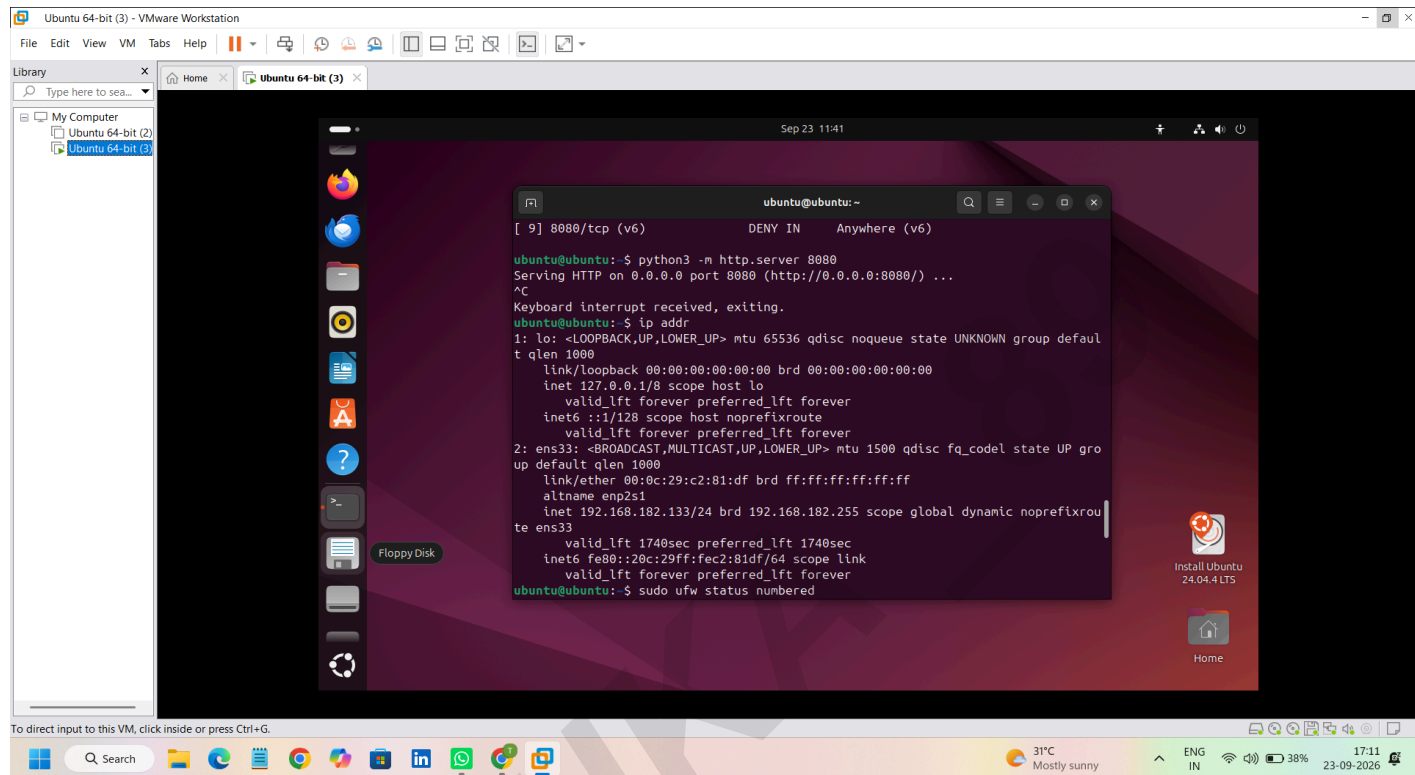
ubuntu@ubuntu:~$
```

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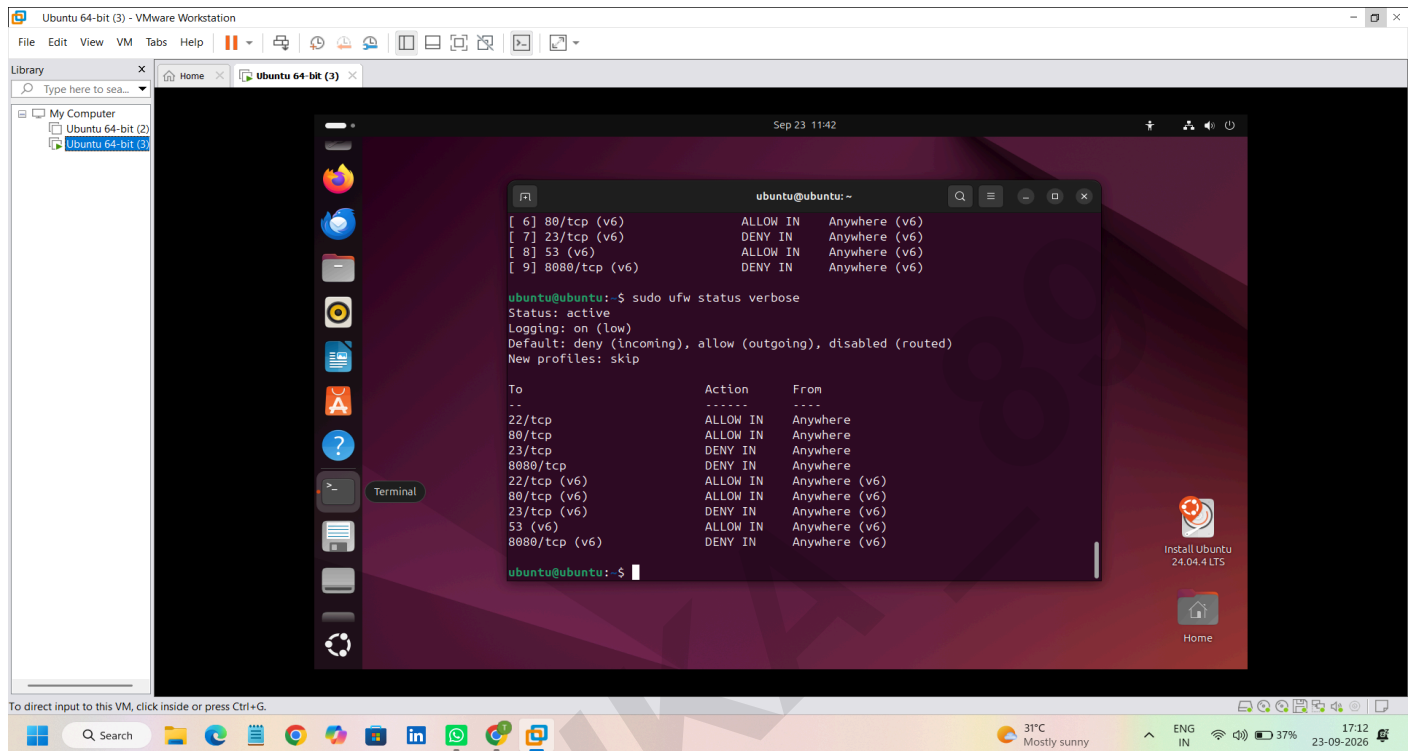


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```
ubuntu@ubuntu: ~  
[ 6] 80/tcp (v6)      ALLOW IN    Anywhere (v6)  
[ 7] 23/tcp (v6)      DENY IN    Anywhere (v6)  
[ 8] 53 (v6)         ALLOW IN    Anywhere (v6)  
[ 9] 8080/tcp (v6)     DENY IN    Anywhere (v6)  
  
ubuntu@ubuntu:~$ sudo ufw status verbose  
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), disabled (routed)  
New profiles: skip  
  
To Action From  
-- -- --  
22/tcp ALLOW IN Anywhere  
80/tcp ALLOW IN Anywhere  
23/tcp DENY IN Anywhere  
8080/tcp DENY IN Anywhere  
22/tcp (v6) ALLOW IN Anywhere (v6)  
80/tcp (v6) ALLOW IN Anywhere (v6)  
23/tcp (v6) DENY IN Anywhere (v6)  
53 (v6) ALLOW IN Anywhere (v6)  
8080/tcp (v6) DENY IN Anywhere (v6)  
  
ubuntu@ubuntu:~$
```

Terminal

Install Ubuntu 24.04.4 LTS

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